Duothan 6.0

Beyond Coding, Into Engineering Excellence.

PHASE 01 · RECON

Assess the damage. Design the recovery.
Prepare the blueprint for rebuilding the future.

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01 · CONTEXT

Scenario Background

A massive global cyber disaster in the year 2065, caused by the Super Malware Agent, has taken down systems across governments, businesses, healthcare, transportation, and financial services. The entire world has been brought to a standstill.

In the financial sector, the damage is severe. Core banking systems, ATMs, digital payment platforms, and loan systems have completely stopped working. Millions of people have lost access to essential financial services. Small businesses can no longer process digital transactions, and people are forced to rely entirely on cash. This loss of secure banking has slowed economic recovery and increased financial inequality across the world.

Fortunately, customer databases are safe because of secure backups, so no user information has been lost. However, the Master Key needed to unlock the banking network is hidden behind strong security layers. While other teams fight the malware in different sectors across the globe, the financial world needs a dedicated rescue effort.

The mission of Duothan 6.0 is to save the digital banking system

Designing, building, and deploying a secure, reliable, and inclusive financial platform from the ground up, using a scalable architecture with independent services.

Rebuilding the old system is not enough. The new platform must use modern cybersecurity, cloud technology, and intelligent automation to make sure an attack like this can never happen again. It needs to restore daily banking operations, protect customer data, support secure transactions, and include a strong disaster recovery plan.

Your mission is to beat the malware, rebuild the financial ecosystem, and restore trusted digital banking for millions of users.

02 · BRIEF

Phase Overview

Your team's first mission is to create a complete project blueprint for rebuilding the digital banking ecosystem destroyed in the 2065 cyber disaster. Rather than writing code immediately, you must analyze the damage, understand user needs, and design a secure solution.

Before starting development, your team must analyze the direct impact of the Super Malware Agent on the financial sector. Your objective is to design a secure, practical banking platform from the ground up - one that solves this crisis, ensures financial services are accessible to everyone, and safely brings trusted digital finance back to the public.

During this phase, you're expected to demonstrate problem analysis, system architecture planning, user experience design, and technical decision-making. Your design must feature a scalable architecture with **independent services**, ensuring a future malware attack can't take down the entire system again.

03 • REQUIREMENTS

Your Tasks

01. Problem Identification

Every successful application begins with identifying a real world problem. Analyze the challenges the digital banking system faced after the Super Malware Agent's attack, and clearly define the specific banking problem your solution will solve.

> *EXPECTED OUTCOME*
> *A clear understanding of the identified banking problem, the affected users, and why restoring a secure and reliable digital banking system is essential for economic recovery.*

02. Proposed Solution

Propose a technology-based solution that addresses the problem you identified. Explain how your banking application will work, how it will securely restore financial services, and the value it provides users.

> *EXPECTED OUTCOME*
> *A clear solution concept that connects directly to the problem and demonstrates practical value for rebuilding digital finance.*

03. System Architecture diagram

Design a scalable architecture built from independent services. Describe how components communicate securely and how data flows through the application without risking a total system failure.

> *EXPECTED OUTCOME*
> *A clear architecture diagram showing how the independent services and databases connect.*

04. Wireframes Design

Create simple interface designs that show how users will interact with the new banking system.

> *EXPECTED OUTCOME*
> *Mid-fidelity or high-fidelity wireframes that represent the planned application interfaces and visual design of the banking system.*

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05. Functional Requirements

Explain what the system should actually do from a user's perspective.

> *EXPECTED OUTCOME*
> *A complete list of required system functionalities to guide development.*

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06. Non-Functional Requirements

Define the quality standards of your application. Since this is a post-cyberattack system, it heavily weighs security, disaster recovery, cloud performance, and reliability heavily.

> *EXPECTED OUTCOME*
> *Quality expectations that ensure the application is highly secure, reliable, and production-ready.*

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07. Technology Stack Selection

Identify the programming languages, databases, and cloud technologies you'll use. Your stack should match the project requirements and support a scalable, **independent-service architecture.**

> *EXPECTED OUTCOME*
> *Justified technology choices, explaining how they help build a secure, attack-proof solution.*

04 · SUBMISSION

Final Deliverables

At the end of Phase 1, your team submits one document containing:

01	Problem Identification
02	Proposed Solution
03	System Architecture Diagram
04	Wireframes
05	Functional Requirements
06	Non-Functional Requirements
07	Technology Stack Description

SUBMISSION FORMAT

The final submission must be a single **Microsoft Word document (.docx)** containing all seven deliverables in order. **PDFs, Google Docs links, or other formats will not be accepted.** For the **wireframes section, embed clear screenshots** directly in the document ***and*** include a working link to the original design file (e.g. Figma, Canva, or Adobe XD) so judges can review it in detail.

Submission Link : duothan.ieeensbm.org/submission

05 · EVALUATION

Mark Allocation

Submissions are evaluated against the criteria below. Judges will assess clarity of thought, technical feasibility, and how well each section supports a secure, scalable banking platform - not just completeness.

CRITERIA	WEIGHT	MARKS
Problem Identification	10%	/ 10
Proposed Solution	20%	/ 20
System Architecture	20%	/ 20
Wireframes Design	15%	/ 15
Functional Requirements	15%	/ 15
Non-Functional Requirements	15%	/ 15
Technology Stack Selection	5%	/ 05
TOTAL	**100%**	**/ 100**

Note: weightings above are indicative and may be adjusted by the judging panel to reflect the depth of analysis shown across sections.

BY THE END OF PHASE 1

You should have a complete project blueprint that clearly defines the problem, the secure solution, the technical architecture, the user experience, and the development strategy required to move on to Phase 2.

Awaken your inner warrior. Rebuild the future. Defend the digital world.

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Questions? **duothan@ieeensbm.org**